28.09.2026

Practical -10 (a)

AIM:

Interneworking with routers in cisco packet tracer.

PROCEDURE:

Step 1 - configuring Router 1

1) Open cisco packet tracer and place one router and 2 PCs
2) Select router and open CLI. Press enter to access command prompt.
3) Enter privileged mode using enable
4) Enter global configuration mode using configure terminal.
5) configure fast ethernet 0/0 with:
   IP address: 192.168.10.1
   Subnet mask: 255.255.255.0
6) Use no shutdown to activate interface
7) configure FastEthernet 0/1
   IP address: 192.168.20.1
   subnet mask: 255.255.255.0
8) Use no shutdown to activate interface.

Step 2 - configure PCs

Device	IP Address	Subnet mask	Default gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

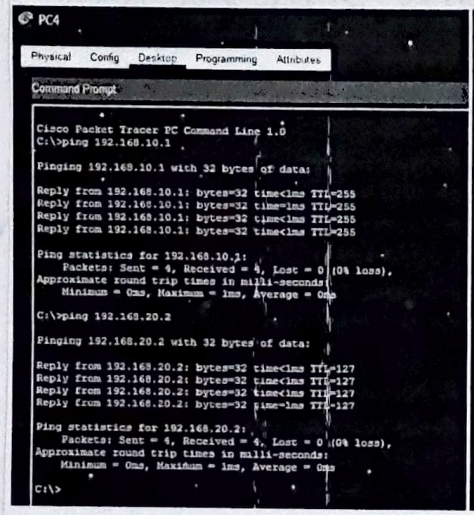
Step -3 - connecting PC's with router.

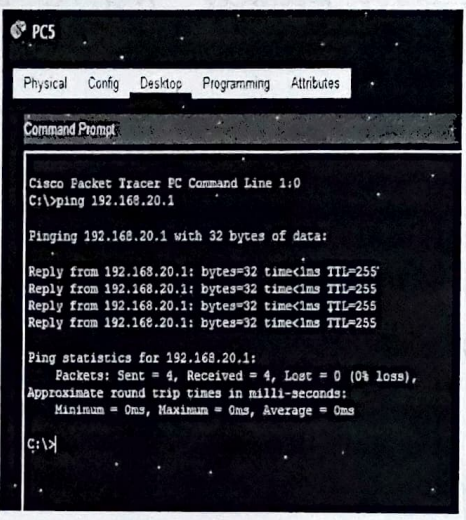
1) Connect PC0 Fast Ethernet to Router1 FastEthernet 0/0 using copper straight-through cable.
2) Connect PC1 Fast Ethernet 0 to Router1 FastEthernet 0/1 using copper cable.
3) Wait until interface become active.
4) Use simulation mode and send a simple PDU from PC0 to PC1
5) Verify that PDU reaches PC1 and acknowledgment is received back to PC0.

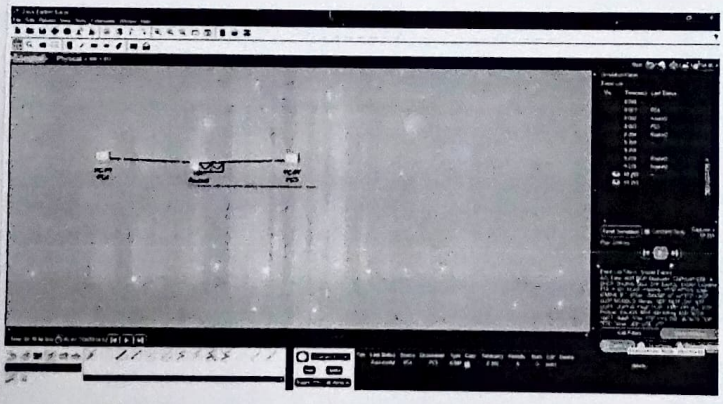
CONFIGURATION TABLE:

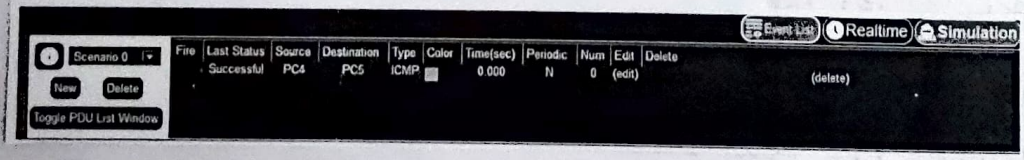
Router configuration:

Device	FastEthernet0/0	subnetmask	Fast Ethernet0/1	subnet mask
Router 1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0









RESULT:

Thus a single internetwork was successfully designed and configured using router in cisco packet tracer.

28.07.2026

Practical -10(b)

AIM:

To design and configure internet using wireless router, DHCP and another cloud.

PROCEDURE:

Step 1 - Build network topology.

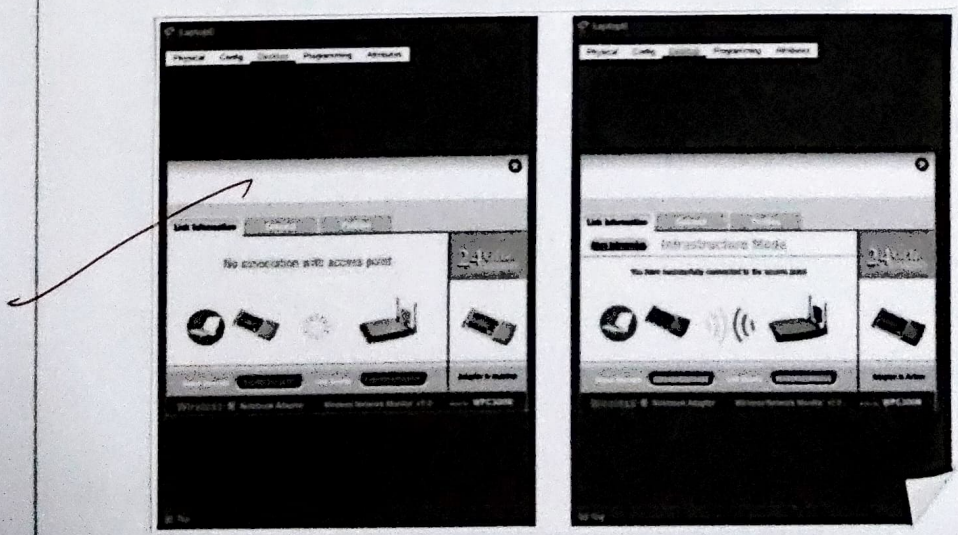
1) Open cisco packet tracer, add devices like PC, Router, cable modem, server
2) connect devices using appropriate cables.

Step 2 - Configure router

1) Wireless router → GUI → Wireless
   Network name → Home Network
2) Go to step tap and ensure DHCP is enabled
3) Configure server as 208.67.220.220
4) Save the settings.

Step 3 - Configure laptop

1) Laptop → physical tab → power off → remove ethernet cable
   add WPC300N wireless module → power on laptop
2) Desktop → PC wireless. select connect tab and select Home network
3) connect laptop to wireless network.



Step 4 - configure PC

1) PC → Desktop → IP configuration - select DHCP
2) PC automatically receives IP address from the wireless routers.
3) use command prompt and ipconfig/all to verify the configuration.

Step 5 - configure internet cloud.

1) open internet cloud
2) config → cable
3) set: From port - coaxial To port: ethernet
4) click Add
5) go to ethernet and select cable as provider n/w

Step 6 - configure cisco.com server

DHCP configuration:

Parameter	value
Pool Name	DHCP pool
Default gateway	208.67.220.220
DNS server	208.67.220.220
Starting IP	208.67.220.1
Subnet mask	255.255.255.0
Maximum user	50

click ADD.

DNS configuration:

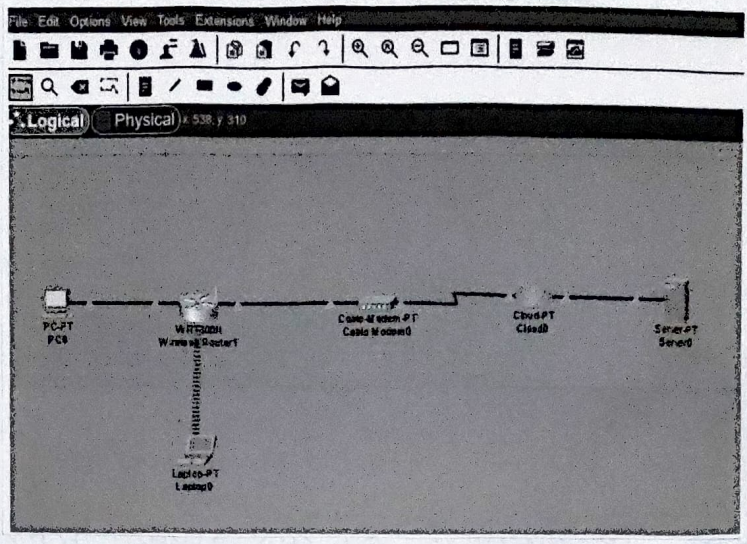
Parameter	Value
Name	cisco.com
Type	A Record
Address	208.67.220.220

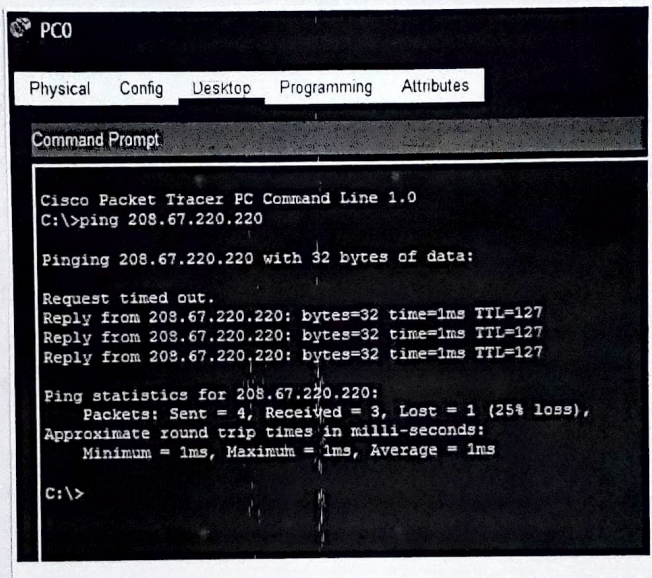
click ADD

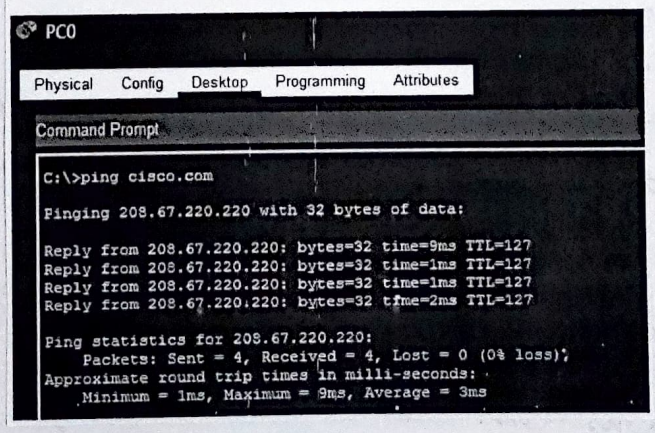
Step -7

Verify connectivity.

✓







RESULT:

Thus the wireless router, DHCP server and internet cloud were successfully configured in cisco packet tracer and connectivity was verified successfully.